

AReplica: User-Centered Design of an Augmented Reality Portion Size Estimation Tool for Dietary Intake Assessment

Nina Rosa*
Cultural Geography
Wageningen University & Research
Travis Masterson
College of Health & Human Development
Pennsylvania State University

Esther Kok, Els Siebelink, Michelle van Alst
Human Nutrition and Health
Wageningen University & Research
Alexander Klippel
Cultural Geography & WANDER XR Experience Lab,
Wageningen University & Research

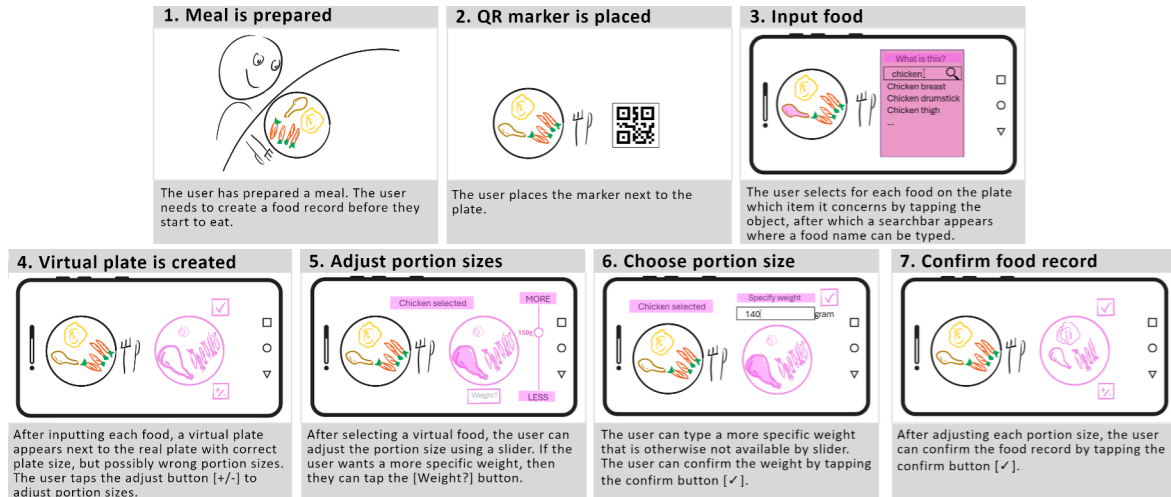


Figure 1: Storyboard illustrating the use of *AReplica* at the beginning of a meal. Pink represent virtual elements; in use the virtual foods are in true colors.

ABSTRACT

Portion size estimation is an essential component of dietary intake assessment, but people generally struggle with such estimation. Augmented reality methods have been developed to aid in this, but it is unclear how selected features are intended to improve accuracy or create a positive user experience. In this paper, we present a storyboard for a new augmented reality approach, *AReplica*. We explain why each feature was chosen and list necessary future research.

Index Terms: Augmented reality, portion size estimation, storyboard

1 INTRODUCTION

Accurate measuring of dietary intake, including energy and nutrients, is an important aspect of nutrition research, and can help gain insight into prevention of non-communicable diseases such as obesity and diabetes. Current validated dietary assessment methods such as food frequency questionnaires, 24-hour dietary recalls, and food diaries have numerous drawbacks that result in inaccurate reporting, e.g. due to recall bias. Moreover, people generally struggle with portion size estimation. Furthermore, this issue has been linked to obesity, suggesting that people with obesity are especially struggling with portion size estimation [2]. Reporting is experi-

enced as tedious and non-engaging, which may also contribute to participant dropout during experiments.

Smartphones have been increasingly used to facilitate intake reporting, not only in terms of mobility and access, but also by off-loading part of the measuring component to a computer. For example, certain automatic systems such as TADA aim to alleviate most user burden by relying on a system to automatically estimate the portion size, based on photographs that the user takes of their meal [4]. However, these systems come with their own drawbacks, such as strict requirements for the potentially multiple images that are required for one meal. For example, a study showed that 12.8% of food images had to be discarded due to errors in the capturing procedure [6].

Another approach to improve portion size estimation is to use augmented reality (AR). AR augments the real world with virtual objects, or in the context of portion size estimation, with a virtual visual aid. While physical visual aids provide a fixed volume for comparison, e.g. a fist, many virtual visual aids provide a shape that can be resized, such that they match the volume of the real food. For example, EatAR lets users fit a virtual dome mesh onto rice [5], The PortionSize app provides users with resizable solid templates (e.g. burger, deck of cards) [1], and the approach by Yang et al. lets users resize a virtual cube wireframe [7]. In each example, the system calculates the corresponding weight or volume.

While these AR approaches have potential to improve estimation accuracy [5] and create a positive user experience, the selected features are chosen without reasoning, and have yet to be compared. We present a new AR tool, *AReplica*, where the main goal is to create a virtual replica of the real meal using a smartphone. As the approach is currently still in design phase, we illustrate it through a

*e-mail: nina.rosa-dejong@wur.nl

storyboard, and explain why certain features were chosen.

2 STORYBOARD

Here we explain how AReplika would be used; see Figure 1 for the storyboard. AReplika is to be used at the beginning of a meal (pane #1), where the user places a marker next to the plate (#2). The user starts the app, and inputs the types of food that are on their plate (#3). Once all foods are selected, a virtual plate appears next to the real plate (#4). This virtual plate is the same size as the real plate, and contains the same foods as the real plate, but possibly with incorrect portion size. If the user wishes to adjust a portion size, they select one of the food components, and tap the adjust button, after which a slider appears with which the user can adjust the portion size (#5). The slider button also shows the corresponding weight in text form. If the user is unsatisfied with the provided portion size options, they can input the weight numerically (#6). This adjustment process can be performed for all food components if necessary. When the user is satisfied with the selected sizes, they can complete the record (#7).

3 DESIGN CHOICES IN AReplika

There are a few design choices here that differ from what has so far been explored in the literature. Firstly, the food selection phase depicted in Figure 1, pane #3 is based on user input. Food identification is not the core of the first AReplika prototype, which will only include limited food selection, but we aim to investigate a combination of automatic food identification and user adjustment at a later stage. Assuming the automatic food identification can be completed in less time than searching, this will shorten the duration of this phase. This can positively affect user experience (e.g. decrease user burden) if the identification is reasonably accurate.

Secondly, previous approaches have either used virtual meshes (e.g. [5]) or solid objects (e.g. [3]). For wire meshes, the intention is typically to have a basic 3D shape such as a sphere or rectangular box approximately fit to the real food shape. Since the object is a mesh, it can be overlaid on the food while the real food can still be seen underneath. Although meals and their presentation vary between cultures, in practice, these basic 3D shapes hardly ever occur at meal time, thus a certain amount of error is to be expected. Yang et al. attempt to circumvent this by just using one shape, namely a cube [7]. However, users required training to keep the error within a reasonable level [7]. Therefore, the foods chosen here are to be presented as realistic, non-transparent food objects, in order to replicate the real food.

The question then arises whether these objects should overlay the real food as would mesh objects. The intention of overlaying is to more easily gauge if the portion sizes match exactly, yet the real food is no longer visible, potentially complicating comparison. Lam et al. compared solid with semi-transparent realistic virtual objects, but found that users' preference depended on their educational background [3]. From a sensory perceptive, the virtual objects will always be presented in front of the real objects, suggesting a difference in depth, which may result in additional error. It is also unlikely that the realistic solid objects will match the real food shape exactly, ruling out the possibility of matching pixels to each other. Therefore, side-by-side comparison is a good alternative. Although there is a concern that both plates may not fit well on a phone screen, our initial tests show that a user is able to capture two side-by-side dinner plates on the camera screen without having to move from their chair.

While resizing of a computer-generated 3D mesh (e.g. a cube) and foods of served as units (e.g. 1 chicken drumstick) can be performed continuously, this is not trivial for all solid, realistic food objects (e.g. 1/2 cup peas vs. 1 cup peas), necessitating discrete portion size options. Lam et al. developed the start of a virtual atlas of portion sizes, with four sizes for each food [3]. For the sake

of accuracy and perceived continuity, we aim to include 15 portion sizes. From informal testing, we saw that using more than 15 steps along the short side of a smartphone makes it difficult to select a desired step rather than the neighboring step. In case the user wants to be more precise, they can input the estimated weight, based on provided corresponding textual weights. This also provides the user with a greater sense of control, potentially increasing satisfaction. We chose a slider in favor of a zoom touch-gesture, because a slider provides a visual reference scale for resizing, resulting in a potentially shorter interaction time. Also, only one finger covers the screen rather than two, with the trade-off being extra required screen space.

Lastly, while the first prototype will present default initial portion sizes (Figure 1, #4), in the future we aim to combine an automatic volume estimator with user adjustment. This has also been suggested by the literature in order to reduce user burden [1].

Most of the design choices were made with accuracy and user experience in mind. We emphasize that each aspect requires testing to understand how accuracy and user experience are influenced in the context of portion size estimation, specifically: mesh versus solid objects, overlay versus side-by-side comparison, resizing by zoom touch-gesture versus slider, user input of food components versus initial food identification with user adjustment, and default size with user adjustment versus initial portion size estimation with user adjustment. To date, these specific aspects have not been addressed in the literature in the context of AR portion size estimation.

4 CONCLUSION

In this paper we presented a storyboard for a new portion size estimation app, AReplika. While design choices of existing AR portions size estimation applications often lacked reasons for selecting certain features, we provided reasons for our design choices from a user experience perspective. We outlined next steps in testing and development.

REFERENCES

- [1] H. E. Diktas, C. P. Lozano, S. Saha, S. T. Broyles, C. K. Martin, and J. W. Apolzan. Evaluating the validity of the portionsize smartphone application for estimating dietary intake in free-living conditions: A pilot study. *Journal of Nutrition Education and Behavior*, 56(9):643–652, 2024. doi: 10.1016/j.jneb.2024.05.226 1, 2
- [2] A. H. Goris, M. S. Westerterp-Plantenga, and K. R. Westerterp. Under-eating and underrecording of habitual food intake in obese men: selective underreporting of fat intake12. *The American Journal of Clinical Nutrition*, 71(1):130–134, 2000. doi: 10.1093/ajcn/71.1.130 1
- [3] M. C. Lam, N. A. Suwadi, A. H. Mohd Zainul Arifien, B. K. Poh, N. S. Safi, and J. E. Wong. An evaluation of a virtual atlas of portion sizes (VAPS) mobile augmented reality for portion size estimation. *Virtual Reality*, 25:695–707, 2021. 2
- [4] Z. Shao, Y. Han, J. He, R. Mao, J. Wright, D. Kerr, C. J. Boushey, and F. Zhu. An integrated system for mobile image-based dietary assessment. In *Proceedings of the 3rd Workshop on AIxFood, AI & Food'21*, p. 19–23. Association for Computing Machinery, New York, NY, USA, 2021. doi: 10.1145/3475725.3483625 1
- [5] T. Stütz, R. Dinic, M. Domhardt, and S. Ginzinger. Can mobile augmented reality systems assist in portion estimation? A user study. In *2014 IEEE International Symposium on Mixed and Augmented Reality - Media, Art, Social Science, Humanities and Design (ISMAR-MASH'D)*, pp. 51–57, 2014. doi: 10.1109/ISMAR-AMH.2014.6935438 1, 2
- [6] M. F. Vasiloglou, K. van der Horst, T. Stathopoulou, M. P. Jaeggi, G. S. Tedde, Y. Lu, and S. Mougiakakou. The human factor in automated image-based nutrition apps: Analysis of common mistakes using the goFOOD Lite App. *JMIR Mhealth Uhealth*, 9(1):e24467, Jan 2021. 1
- [7] Y. Yang, W. Jia, T. Bucher, H. Zhang, and M. Sun. Image-based food portion size estimation using a smartphone without a fiducial marker. *Public Health Nutrition*, 22(7):1180–1192, 2019. doi: 10.1017/S136898001800054X 1, 2